📕 Developer Manual — Life Vest Tracker

📋 Daftar Isi

1. Tech Stack 3
2. Setup Development 3
    - Prasyarat 3
    - Langkah Instalasi 3
3. Struktur Folder 4
4. Database Schema 6
    - Tabel aircraft 6
    - Tabel seats 6
    - Tabel airlines 7
    - Relasi 7
5. Routes & Controllers 7
    - Routes (routes/web.php) 7
    - Controllers 8
        - DashboardController 8
        - AircraftController 9
        - ReportController 9
        - FleetController 9
6. Membuat Layout Baru (Step-by-Step) 9
    - Step 1: Buat Partial (Seat Map) 9
    - Step 2: Buat Template Wrapper 12
    - Step 3: Konfigurasi Class Rows 13
    - Step 4: Auto-Detect 13
7. Menambah Tipe Pesawat Baru 13
    - 1. Buat Layout 13
    - 2. Update Buffer Spare di Blank Form 14
8. Menambah Airline Baru 14
    - Via Fleet Manager (UI) 14
    - Via Migration/Seeder (Code) 14
    - Via Migration 14
9. CSS & Theming 15
    - File CSS 15
    - Dark / Light Mode 15
    - Kelas Status 15
10. Sistem PDF 16
    - Library 16
    - Template PDF 16
    - Cara Kerja 16

Penting untuk DomPDF..... 16
Spare Buffer Blank Form..... 16
11. Alur Request..... 17
Alur Update Tanggal Expiry (Seat Map)..... 17
Alur Batch Input..... 17
Alur Export PDF..... 17

1. 🛠️ Tech Stack

Komponen	Teknologi
Backend	Laravel 12 (PHP 8.1+)
Frontend	Vanilla CSS & JavaScript
Database	MySQL / MariaDB
Build Tool	Vite
PDF	Barryvdh/DomPDF
Timezone	Asia/Jakarta (GMT+7)

2. 🚀 Setup Development

Prasyarat

- PHP 8.1+
- Composer
- Node.js & npm
- MySQL/MariaDB (via Laragon/XAMPP)

Langkah Instalasi

```
# 1. Clone & install
git clone <repo-url>
cd lifevest-laravel
composer install
npm install

# 2. Setup environment
cp .env.example .env
php artisan key:generate

# 3. Konfigurasi database di .env
DB_DATABASE=lifevest_tracker
DB_USERNAME=root
DB_PASSWORD=
```

```
# 4. Migrasi database
php artisan migrate

# 5. (Opsional) Seed data awal
php artisan db:seed

# 6. Jalankan server (buka 2 terminal)
# Terminal 1:
php artisan serve

# Terminal 2:
npm run dev
```

Buka http://localhost:8000

3. 📁 Struktur Folder

```
lifevest-laravel/
├── config/
│   └── aircraft_class_rows.php      # Konfigurasi baris per class per layout
├── database/
│   ├── migrations/
│   │   ├── create_aircraft_table    # Tabel pesawat
│   │   ├── create_seats_table       # Tabel kursi & expiry
│   │   └── create_airlines_table    # Tabel maskapai
│   └── seeders/
│       ├── AircraftSeeder.php       # Data awal pesawat
│       └── AirlineSeeder.php        # Data awal airline
├── app/
│   ├── Models/
│   │   ├── Aircraft.php             # belongsTo(Airline)
│   │   ├── Airline.php              # hasMany(Aircraft)
│   │   └── Seat.php                 # Data kursi & expiry date
│   └── Http/Controllers/
│       ├── DashboardController.php  # Dashboard & Fleet Overview
│       ├── AircraftController.php   # Seat map, update, batch input
│       ├── FleetController.php      # CRUD pesawat & airline
│       └── ReportController.php     # PDF Export & Blank Form
├── resources/
│   ├── views/
│   │   ├── layouts/app.blade.php    # Master layout + Navbar
│   │   ├── dashboard.blade.php      # Halaman dashboard
```

```
│   │   ├── fleet/              # Fleet Manager views
│   │   │   ├── index.blade.php      # Daftar pesawat & airline
│   │   │   ├── create.blade.php     # Form tambah pesawat
│   │   │   └── edit.blade.php       # Form edit pesawat
│   │   ├── aircraft/           # ⭐ Template wrapper per layout
│   │   │   ├── b737-e46.blade.php   # Wrapper B737 layout e46
│   │   │   ├── b777-3class.blade.php # Wrapper B777 3-class
│   │   │   └── ... (16 file)
│   │   ├── aircraft/partials/      # ⭐ Seat map layout (inti)
│   │   │   ├── b737-e46.blade.php   # Konfigurasi kursi B737
│   │   │   ├── b777-3class.blade.php # Konfigurasi kursi B777
│   │   │   └── ... (16 file)
│   │   ├── reports/            # Template PDF
│   │   │   ├── seat-map.blade.php   # PDF Export (berwarna)
│   │   │   └── blank-form.blade.php  # Blank Form (kosong)
│   │   └── components/         # Blade components
│   │       ├── toolbar.blade.php    # Toolbar + tombol Export
│   │       ├── seat-cell.blade.php  # Komponen satu sel kursi
│   │       ├── cockpit-section.blade.php
│   │       ├── status-legend.blade.php
│   │       ├── aircraft-header-info.blade.php
│   │       └── date-modal.blade.php
│   ├── css/
│   │   ├── style.css           # CSS utama + Dark/Light mode
│   │   └── dashboard.css       # CSS khusus dashboard
│   └── js/
│       └── app.js              # JavaScript interaksi kursi
└── routes/
    └── web.php                 # Semua route
```

4. 🗄️ Database Schema

Tabel aircraft

Kolom	Tipe	Keterangan
id	int (PK)	Auto increment
registration	varchar (unique)	Nomor registrasi (PK-GFD)
type	varchar	Tipe pesawat (B737-800)
icon	varchar	Emoji ikon (default: ✈️)
layout	varchar	Kode layout blade (b737-e46)
status	enum	active / prolong
airline_id	int (FK)	Foreign key ke tabel airlines

Tabel seats

Unique constraint: (registration, seat_id)

Kolom	Tipe	Keterangan
id	int (PK)	Auto increment
registration	varchar	FK ke aircraft registration
seat_id	varchar	ID kursi: 21A, captain, pax-1, inf-3
row	int (nullable)	Nomor baris (null untuk cockpit)
col	varchar (nullable)	Huruf kolom (A, B, C, ...)
class_type	varchar	cockpit, first, business, economy
expiry_date	date (nullable)	Tanggal kadaluarsa life vest

Tabel airlines

Kolom	Tipe	Keterangan
id	int (PK)	Auto increment
name	varchar	Nama maskapai
code	varchar (nullable)	Kode IATA (GA, QG)
icon	varchar	Emoji (default: 🏢)

Relasi

airlines (1) ——— (N) aircraft
aircraft.registration (1) ——— (N) seats

5. 🛤️ Routes & Controllers

Routes (routes/web.php)

Method	URL	Controller/Method	Name	Deskripsi
GET	/	DashboardController@__invoke	dashboard	Halaman dashboard
GET	/aircraft/{reg}	AircraftController@show	aircraft.show	Seat map
POST	/aircraft/{reg}/update-seats	AircraftController@updateSeats	aircraft.updateSeats	Update expiry
DELETE	/aircraft/{reg}/delete-seat	AircraftController@deleteSeat	aircraft.deleteSeat	Hapus spare seat
GET	/aircraft/{reg}/report	ReportController@exportPdf	reports.pdf	Export PDF

GET	/aircraft/{reg}/blank-form	ReportController@exportBlankForm	reports.blank	Blank Form
GET	/aircraft/{reg}/batch-input	AircraftController@batchInput	aircraft.batchInput	Batch Input form
POST	/aircraft/{reg}/batch-input	AircraftController@storeBatchInput	aircraft.storeBatchInput	Simpan batch
Resource	/fleet	FleetController	fleet.*	CRUD pesawat
GET	/fleet/airlines/create	FleetController@createAirline	airlines.create	Form tambah airline
POST	/fleet/airlines	FleetController@storeAirline	airlines.store	Simpan airline
GET	/fleet/airlines/{id}/edit	FleetController@editAirline	airlines.edit	Form edit airline
PUT	/fleet/airlines/{id}	FleetController@updateAirline	airlines.update	Update airline
DELETE	/fleet/airlines/{id}	FleetController@destroyAirline	airlines.destroy	Hapus airline

Controllers

DashboardController

- Menghitung status semua seat per pesawat
- Mengelompokkan per airline → per tipe
- Menghitung $perFleetStats untuk dropdown filter fleet
- Menghitung P/N summary

AircraftController

- show() — Memuat seat map; return view aircraft/{layout}.
- updateSeats() — Update tanggal expiry untuk satu atau banyak kursi.
- deleteSeat() — Hapus spare seat (PAX/INF) dari database.
- batchInput() — Render form batch input dengan sections berdasarkan config/aircraft_class_rows.php.
- storeBatchInput() — Proses input massal; membuat/update record Seat per kolom per baris.
- parseFlexibleDate() — Parse berbagai format tanggal (Oct-25, 24-Jan-25, 01/03/2030).

ReportController

- exportPdf() — Render PDF lengkap dengan warna status dan tanggal.
- exportBlankForm() — Render PDF form kosong; menentukan maxPax/maxInf berdasarkan tipe pesawat.

FleetController

- Standard Laravel Resource Controller untuk CRUD pesawat.
- Tambahan method untuk CRUD airline (createAirline, storeAirline, dll).

6. ✈️ Membuat Layout Baru (Step-by-Step)

Jika ada pesawat dengan konfigurasi kursi yang belum pernah ada, ikuti langkah berikut:

Step 1: Buat Partial (Seat Map)

File ini berisi konfigurasi kursi sesungguhnya.

Lokasi: resources/views/aircraft/partials/

1. Copy file yang paling mirip. Contoh untuk A330 layout baru:

```
cp resources/views/aircraft/partials/a330-300a.blade.php
resources/views/aircraft/partials/a330-300d.blade.php
```

2. Edit file baru tersebut. Struktur partial:

```
<!-- Cabin Section -->
<section class="cabin-section">
    <h2>Economy</h2>
    <div class="cabin-grid">
        <!-- Header kolom -->
        <div class="header-row">
            <div class="row-number"></div>
            <div class="col-header" data-col="A">A</div>
```

```
            <div class="col-header" data-col="B">B</div>
            <div class="col-header" data-col="C">C</div>
            <div class="aisle"></div>
            <div class="col-header" data-col="D">D</div>
            <div class="col-header" data-col="E">E</div>
            <div class="col-header" data-col="F">F</div>
        </div>

        <!-- Baris kursi -->
        @foreach(range(21, 46) as $row)
            @if(!in_array($row, [13, 24])) {{-- Skip row --}}
            <div class="seat-row" data-row="{{ $row }}">
                <div class="row-number" data-row="{{ $row }}">{{ $row
}}</div>
                @foreach(['A','B','C'] as $col)
                    <x-seat-cell :row="$row" :col="$col" :seats="$seats" />
                @endforeach
                <div class="aisle"></div>
                @foreach(['D','E','F'] as $col)
                    <x-seat-cell :row="$row" :col="$col" :seats="$seats" />
                @endforeach
            </div>
            @endif
        @endforeach
    </div>
</section>

<!-- Spare Section: PAX & INF -->
<section class="cabin-section">
    <h2>Spare</h2>
    <div class="spare-grid">
        <!-- PAX Column -->
        <div class="spare-column" id="pax-column">
            <h3>PAX</h3>
            <div class="spare-items" id="pax-items">
                @php
                    $paxSeats = collect($seats)->filter(fn($s, $id) =>
str_starts_with($id, 'pax-'))
                        ->sortBy(fn($s, $id) => (int) str_replace('pax-',
'', $id));
                @endphp
                @forelse($paxSeats as $seatId => $seat)
```

```
                    @php
                        $num = str_replace('pax-', '', $seatId);
                        $status = $seat?->status ?? 'no-data';
                        $expiryDate =
$seat?->expiry_date?->format($dateFormat) ?? '-';
                    @endphp
                    <div class="seat-card spare-card status-{{ $status }}"
data-seat="{{ $seatId }}">
                        <button type="button" class="btn-delete-spare"
title="Delete">×</button>
                        <div class="seat-id">{{ $num }}</div>
                        <div class="seat-date" data-date="{{
$seat?->expiry_date?->format('Y-m-d') ?? '' }}">
                            {{ $expiryDate }}
                        </div>
                    </div>
                @empty
                    <p class="empty-message">Belum ada data PAX</p>
                @endforelse
            </div>
            <button type="button" class="btn btn-add-spare"
data-type="pax">+ Add PAX</button>
        </div>

        <!-- INF Column (sama structurenya, ganti pax → inf) -->
        ...
    </div>
</section>
```

3. **Sesuaikan**:
   - Kolom header (A–F untuk narrow body, A–K untuk wide body)
   - Range baris (range(21, 46))
   - Konfigurasi aisle (lorong di tengah)
   - Jika ada baris yang di-skip, gunakan: @if(!in_array($row, [13, 24]))

Step 2: Buat Template Wrapper

File ini membungkus partial dengan layout utama.

Lokasi: resources/views/aircraft/

1. Copy template yang mirip:

```
cp resources/views/aircraft/a330-300a.blade.php
resources/views/aircraft/a330-300d.blade.php
```

2. Edit baris @include agar menunjuk ke partial yang baru:

```
@extends('layouts.app')

@section('header-right')
    <x-aircraft-header-info :aircraft="$aircraft"
:registration="$registration" />
@endsection

@section('content')
    <!-- Toolbar -->
    <x-toolbar :registration="$registration" />

    <!-- Part Number Info -->
    <x-part-number-bar :aircraft="$aircraft" :qtyAdult="$qtyAdult ?? 0"
        :qtyCrew="$qtyCrew ?? 0" :qtyInfant="$qtyInfant ?? 0"
        :expAdult="$expAdult ?? 0" :expCrew="$expCrew ?? 0"
:expInfant="$expInfant ?? 0" />

    <!-- Status Legend -->
    <x-status-legend />
    @include('aircraft.partials.a330-300d')  {{-- ← GANTI NAMA INI --}}

    <!-- Date Modal -->
    @include('components.date-modal')
@endsection

@push('scripts')
    <script>
        window.AIRCRAFT_CONFIG = {
            registration: '{{ $registration }}',
            updateUrl: '{{ route('aircraft.updateSeats', $registration)
```

```
}}',
            deleteUrl: '{{ route('aircraft.deleteSeat', $registration) }}',
            csrfToken: '{{ csrf_token() }}'
        };
    </script>
@endpush
```

Step 3: Konfigurasi Class Rows

Edit config/aircraft_class_rows.php:

```
'a330-300d' => [
    'business' => range(6, 11),
    'economy'  => array_diff(range(21, 55), [24]), // skip row 24
],
```

Penting: Konfigurasi ini menentukan class_type saat menyimpan seat ke database.

Step 4: Auto-Detect

Selesai! Sistem akan **otomatis mendeteksi** layout baru.

Saat menambahkan pesawat di Fleet Manager, pilihan a330-300d akan langsung muncul di dropdown layout.

> ⚠️ **Catatan Penting**: Nama file partial dan template **HARUS SAMA** (contoh: kedua file bernama a330-300d.blade.php).

7. ➕ Menambah Tipe Pesawat Baru

Jika menambahkan **tipe pesawat yang belum ada** (misalnya Boeing 787), lakukan:

1. Buat Layout

Ikuti Step 1–4 di atas.

2. Update Buffer Spare di Blank Form

Edit app/Http/Controllers/ReportController.php, method exportBlankForm():

```
// Tambahkan elseif baru:
} elseif (str_contains($type, '787')) {
    $maxPax = 20; $maxInf = 30;  // Sesuaikan angka buffer
}
```

3. Tambah Pesawat via Fleet Manager

Buka /fleet → Add New Aircraft → Pilih layout yang baru dibuat.

8. 🏢 Menambah Airline Baru

Via Fleet Manager (UI)

1. Buka /fleet → Tab Airlines → **+ Add New Airline**
2. Isi nama dan kode IATA → Save

Via Migration/Seeder (Code)

```
// database/seeders/AirlineSeeder.php
DB::table('airlines')->insert([
    'name' => 'Batik Air',
    'code' => 'ID',
    'icon' => '🏢',
    'created_at' => now(),
    'updated_at' => now(),
]);
```

Via Migration

```
// Di migration create_airlines_table.php sudah ada insert default:
DB::table('airlines')->insert([
    ['name' => 'Garuda Indonesia', 'code' => 'GA', 'icon' => '🦅',
'created_at' => now(), 'updated_at' => now()],
]);
```

9. 🎨 CSS & Theming

File CSS

File	Fungsi
resources/css/style.css	CSS utama (seat map, toolbar, fleet, dll)
resources/css/dashboard.css	CSS khusus dashboard

Dark / Light Mode

Tema menggunakan CSS Variables di :root dan [data-theme="dark"]:

```
:root {
    --bg-primary: #15161a;
    --bg-secondary: #1e1f25;
    --bg-card: #25262d;
    --text-primary: #ffffff;
    --text-secondary: #9ca3af;
    --primary: #3b82f6;
    --danger: #ef4444;
    --border: #2d2e36;
    /* ... dll */
}
```

Toggle tema disimpan di localStorage oleh JavaScript di layouts/app.blade.php.

Kelas Status

Warna status life vest:

```
.status-safe     { border-color: #22c55e; background: rgba(34, 197, 94, 0.15); }
.status-warning  { border-color: #f59e0b; background: rgba(245, 158, 11, 0.15); }
.status-critical { border-color: #ef4444; background: rgba(239, 68, 68, 0.15); }
.status-expired  { border-color: #8b5cf6; background: rgba(139, 92, 246, 0.15); }
.status-no-data  { border-color: #64748b; background: rgba(100, 116, 139, 0.1); }
```

10. 📄 Sistem PDF

Library

Menggunakan **Barryvdh/DomPDF** (barryvdh/laravel-dompdf).

Template PDF

File	Fungsi
reports/seat-map.blade.php	PDF lengkap (dengan warna & tanggal)
reports/blank-form.blade.php	PDF form kosong (untuk inspeksi manual)

Cara Kerja

1. Controller memanggil Pdf::loadView('reports.xxx', $data)
2. View blade di-render menjadi HTML
3. DomPDF mengkonversi HTML → PDF
4. PDF di-stream ke browser ($pdf->stream(...))

Penting untuk DomPDF

- CSS **inline** atau <style> tag di dalam view (tidak bisa external CSS).
- **Tidak support** semua CSS modern (contoh: :last-of-type, flexbox terbatas, grid tidak didukung).
- Gunakan display: inline-block dan float untuk layout.
- page-break-inside: avoid untuk menghindari potongan bagian.
- Ukuran kertas diset via $pdf->setPaper('a4', 'portrait').

Spare Buffer Blank Form

Di ReportController::exportBlankForm(), jumlah kotak spare ditentukan per tipe:

```
if (str_contains($type, 'A320')) {
    $maxPax = 15; $maxInf = 20;
} elseif (str_contains($type, 'A330')) {
    $maxPax = 15; $maxInf = 40;
} elseif (str_contains($type, 'ATR')) {
    $maxPax = 10; $maxInf = 10;
} elseif (str_contains($type, '737')) {
    $maxPax = 10; $maxInf = 25;
} elseif (str_contains($type, '777')) {
    $maxPax = 35; $maxInf = 40;
}
```

Nilai ini ditentukan dengan menambahkan buffer di atas jumlah maksimum spare aktual.

11. 🔄 Alur Request

Alur Update Tanggal Expiry (Seat Map)

```
User klik kursi → Select kursi → Klik "Set Date" → Pilih tanggal → Klik "Apply"
  ↓
JavaScript (app.js) → POST /aircraft/{reg}/update-seats
  ↓
AircraftController@updateSeats
  ↓
Seat::updateOrCreate() → Update database
  ↓
Response JSON → JavaScript update warna kursi di UI
```

Alur Batch Input

```
User buka Batch Input → Paste data dari Excel → Klik "Save All"
  ↓
POST /aircraft/{reg}/batch-input
  ↓
AircraftController@storeBatchInput
  ↓
Loop per section → per kolom → per baris:
  parseFlexibleDate() → Seat::updateOrCreate()
  ↓
Redirect ke seat map dengan pesan sukses
```

Alur Export PDF

```
User klik "Export PDF" → GET /aircraft/{reg}/report
  ↓
ReportController@exportPdf
  ↓
Seat::where(registration) → keyBy(seat_id)
  ↓
Pdf::loadView('reports.seat-map', $data)
  ↓
DomPDF render HTML → PDF
  ↓
$pdf->stream() → Browser menampilkan PDF
```